

Sudoku Game (10-Week Plan)

Week 1

- 1.project topic selection

Week 2 : project topic selection and planning

- 1.project overview
- 2.project objective
- 3.Sudoku game introduction
- 4.Project scope
- 5.Basic features
- 6.Tools selection

Week 3 : design the game interface

- 1.Sudoku game flowchart diagram
- 2.Simple ui layout design

Week 4 : Tools setup and environment preparation

1. Create the Sudoku game window.
- 2.Create a 4×4 Sudoku grid.
- 3.Add input cells for entering numbers.
- 4.Add basic buttons such as New Game and Clear.
- 5.Arrange the components properly.
- 6.Run and check the basic interface in NetBeans.

Week 5 : Basic grid implementation

- 1.Create the 4×4 Sudoku grid.
- 2.Allow the player to enter numbers into empty cells.
- 3.Set some cells with fixed numbers.
- 4.Add basic number validation.
- 5.Test whether numbers can be entered correctly.
- 6.Run and check the game in NetBeans.

Week 6 : Game control logic

- 1.Add number validation.
- 2.Check duplicate numbers in rows.
- 3.Check duplicate numbers in columns.
- 4.Check duplicate numbers in each 2×2 box.
- 5.Show an error message for incorrect entries.
- 6.Test the game logic

Week 7 : Number filling system and puzzle setup

- 1.Create a sample 4×4 Sudoku puzzle.
- 2.Set the fixed numbers in the grid.
- 3.Keep some cells empty for the player.
- 4.Add a New Game option.
- 5.Test different puzzle arrangements.

Week 8 : Player input system and game features

1. Allow players to enter numbers in empty cells.
- 2.Prevent changes to fixed cells.
- 3.Add a Clear button to remove entered numbers.
- 4.Check player inputs for valid numbers.
- 5.Display messages for incorrect inputs.
- 6.Test the input features.

Week 9 : Perform debugging , error fixing and final testing

1. Add a Winning Message when the puzzle is solved.
- 2.Add a Restart Game option.
- 3.Improve the game interface.
- 4.Add simple instructions for players.
- 5.Check and improve the overall game performance.

Week 10 : prepare project documentation and final project submission

- 1.Documentation
- 2.Screenshots to the final game
- 4.Conclusion
- 5.complete the project report
- 6.Final submission